

1. (a) Explain Card and Krueger's seminal study regarding minimum wage and employment by using the difference-in-differences (DID).
(b) Explain the linear-log model.
2. Suppose that in 2021 a city with two flagship universities was given a sharp increase in funding. Call the two universities Eastern and Western, respectively. The local government decided that the funding would go to Western, mostly to hire more teachers to reduce class sizes. To evaluate the effectiveness of the increased funding, the government collected random samples of the students in 2020 and 2022 at both universities. The measure of student performance is a standardized test score given to all undergraduate students in the city.
 - (a) Let *score* be the test score. Without controlling for other factors, write down a difference-in-differences (DID) model using a linear regression with an interaction term that allows you to determine whether the additional funding led to improved test scores. Define precisely all the regressors (explanatory variables) in your model. Explain the specific meaning of all the parameters in the model.
 - (b) Illustrate your DID model by using a diagram. What is the key assumption for your model? Explain.
 - (c) How might estimating the model in (a) be biased if Western gets an influx of better students because of the increased funding? Explain.
3. Consider the following two specifications of regression models:

$$Y = \beta_0 + \beta_1 Treat + \beta_2 Treat \cdot After + u, \quad (1)$$

$$Y = \beta_0 + \beta_1 After + \beta_2 Treat \cdot After + u, \quad (2)$$

where *Treat* is a dummy variable that equals 1 for Treatment Group and equals 0 otherwise; *After* is a dummy variable that equals 1 for the time period after the treatment/policy change and equals 0 otherwise.

- (a) Explain the specific meaning of the parameters β_0 , β_1 , and β_2 in the model (1). Illustrate the model by using a diagram.
- (b) Explain the specific meaning of the parameters β_0 , β_1 , and β_2 in the model (2). Illustrate the model by using a diagram.

- (c) Compare the two above models with a difference-in-differences model. Which model is better? Explain.
4. Acemoglu, Johnson, and Robinson (AJR, 2024 Nobel Laureates in Economics) are interested in the effect of good political institutions (e.g., strong rule-of-law, protection of property rights) on prosperity and long-term economic performance. They used a sample of 64 former European colonies and argue that European colonizing countries tended to set up colonies as either an “extractive state” or as a “migrant colony”.

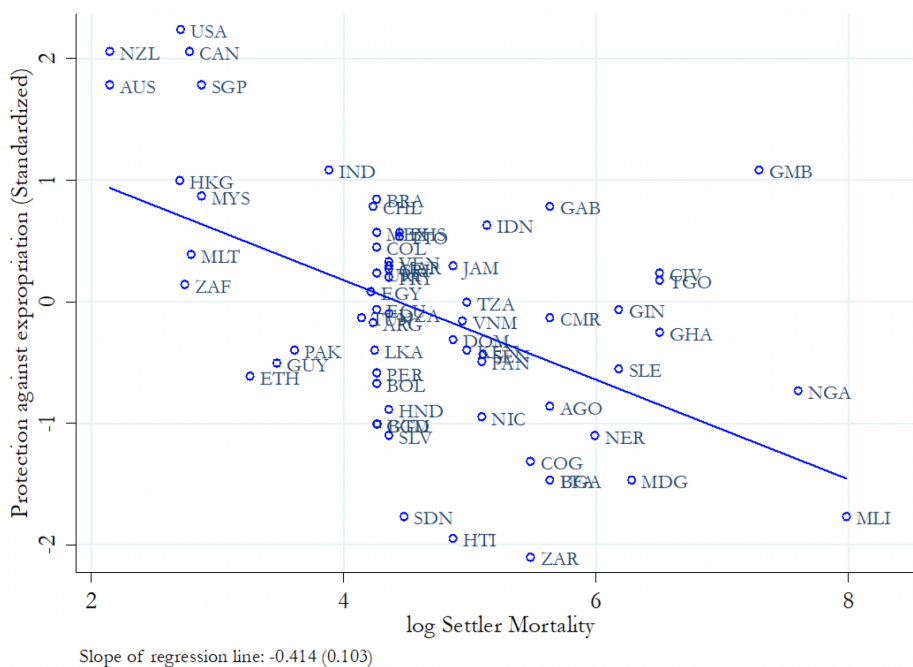
AJR highlighted that initial conditions in areas colonized by Europeans differed largely by disease environments, in particular for the settlers themselves. They hypothesized that Europeans settled to a greater extent in places where their mortality rates were relatively low; they brought along their values as well as beliefs and developed home-like institutions, in particular economic and political institutions benefitting themselves. In areas where mortality was high due to diseases such as malaria and yellow fever, Europeans did not settle in large numbers. In these places, therefore, the colonizers instead established or maintained institutions designed to extract resources from native populations as much and as quickly as possible. Many characteristics of the colonial state and institutions persisted even after independence, and that such persistence influences contemporary economic performance.

Figure 2: Proposed causal chain in Acemoglu, Johnson, and Robinson (2001)

1. Living conditions during colonization (settler mortality) $\Rightarrow$ 2. Size of European settlements during colonization $\Rightarrow$ 3. Colonial institutions $\Rightarrow$ 4. Institutions at independence $\Rightarrow$ 5. Contemporary institutions $\Rightarrow$ 6. Contemporary economic prosperity

- (a) Discuss which countries in Panel B are more likely to be “extractive state” and “migrant colony”, respectively.
- (b) Construct a linear-log model for “Protection Against Expropriation Risk” (1985-95) and “Settler Mortality” (19th century), and explain the meaning of the slope estimate -0.414.

Panel B



Notes. Panel A: Relationship between log GDP per capita in 1995 (purchasing power adjusted) and log of settler mortality and. Panel B: Relationship between average protection against expropriation risk 1985-95 and log of settler mortality. Panels A and B are analogous to Figures 1 and 3 in Acemoglu, Johnson, and Robinson (2001). They have been reproduced using data from Acemoglu, Johnson, and Robinson (2012).

5. Empirical Exercise: Use the dataset **lawsch_ranking.dta** uploaded in NTU Learn. The variable *llibvol* is equal to $\log(\text{libvol})$, the log of number of volumes in library, and *lcost* is equal to $\log(\text{cost})$, the log of law school cost such as tuition fee.
 - (a) Run the regression of *salary* (the starting salary of law school graduates) on *top10*, *r11_25*, *r26_40*, *r41_60*, *GPA*, *llibvol* and *lcost*. Interpret the estimated coefficients of *llibvol* and *lcost*.
 - (b) Test the hypothesis that ranking does not matter for the starting salary using both homoskedasticity-only and heteroskedasticity-robust *F*-tests.